

tained the optimal power allocation to maximize the secrecy rate. In the partial CSI case, we derived a closed-form expression of SOP and solved the SRM problem under the SOP constraint to get a near-optimal power allocation. Moreover, we give a scheme with one chosen helper in the partial CSI case, which can achieve satisfactory performance in some scenarios and reduce the complexity dramatically. The SUCJ scheme can achieve significant performance improvements compared with SISO and RO schemes, and it is more applicable than the other existing schemes for enhancing the physical layer security of the uplink communication of cellular systems with uncoordinated single-antenna mobile stations.

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Radio Environment Map-Aided Doppler Shift Estimation in LTE Railway

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Abstract—Due to the high mobility of high-speed trains (HSTs), Doppler shift estimation has been a big challenge for HSTs. In this paper, we consider an orthogonal frequency-division multiplexing (OFDM) system based on the long-term evolution (LTE) railway standard and design the novel Doppler shift estimation algorithm. By exploiting features of HSTs, i.e., regular and repetitive routes and timetables, resulting in a predictable Doppler shift curve, a radio environment map (REM) including the Doppler shift information can be constructed via field tests. Based on REM, a maximum *a posteriori* estimator (MAPE) is proposed to provide an accurate estimation of Doppler shift. It uses the estimation from REM (REME) as *a priori* knowledge and exploits the cyclic prefix (CP) structure of OFDM to provide a maximum *a posteriori* estimation. The Cramer–Rao lower bounds (CRLBs) are derived. The performance of MAPE is evaluated via simulations and compared to that of REME, the classical CP-based estimator, and other existing methods. It is shown that MAPE significantly outperforms the existing methods in terms of both estimation mean square error (MSE) and bit error rate.

Index Terms—Doppler shift estimation, high-speed railway, long-term evolution (LTE), maximum *a posteriori* (MAP), orthogonal frequency division multiplexing (OFDM).

I. INTRODUCTION

Recently, high-speed trains (HSTs) have been widely deployed worldwide, due to their various advantages such as high speed and high energy efficiency. To provide a safe and fast trip to passengers, advanced train controls like on-train video monitoring should be employed in HSTs. Moreover, passengers also need high-speed wireless access for businesses or entertainments during the trip. However, the current narrowband wireless communication system for HSTs cannot meet these demands. It is, thus, necessary to develop broadband wireless communication systems for HSTs. In this regard, the international

Manuscript received June 1, 2015; revised October 2, 2015 and June 23, 2016; accepted July 25, 2016. Date of publication August 11, 2016; date of current version May 12, 2017. This work was supported by Beijing Talents fund 2015000021223ZK31 and the National Natural Science Foundation of China under Grant 61571425, by the Australian Research Council under Grant DP150104019 and Grant FT120100487, and by the Faculty of Engineering and Information Technologies, The University of Sydney, under the Faculty Research Cluster Program. The work of Z. Hou was also supported by International Postgraduate Research Scholarship and Australian Postgraduate Award. The review of this paper was coordinated by Dr. D.-W. Matolak. (*Corresponding author: Yiqing Zhou.*)

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Digital Object Identifier 10.1109/TVT.2016.2599558

union of railways has recommended to employ long-term evolution (LTE) based on the fourth-generation (4G) technology for HSTs, which is referred to LTE-Railway (LTE-R).

Due to the high velocity of HSTs, the Doppler shift in HST is large (e.g., 1350 Hz) and rapidly varying with time (e.g., 4.9 kHz/s in current systems when the train passes by a base station) [1]. Such a high Doppler shift will be detrimental to the orthogonal frequency division multiplexing (OFDM)-based LTE-R system, which is very sensitive to large Doppler shift. Various schemes have been proposed for Doppler shift estimation in high-mobility scenarios. For example, the state-space-based Kalman–Bucy filter estimator (KBFE) [2] works well when the Doppler shift changes in the shape of a sine curve. However, in HSTs, the Doppler shift curve is nearly rectangular [1]. Therefore, it would be difficult for KBFE to track the fast changing rectangular curve. Moreover, the line-of-sight (LOS) path-separated estimator (LPSE) [3] is designed for the practical HST channel. It separates the dominant LOS path from other multipaths and uses the LOS path only to conduct the Doppler shift estimation. However, its performance is sensitive to the estimation error of the channel length; thus, it degrades significantly in the low-signal-to-noise ratio (SNR) range. Most of the existing Doppler shift estimation methods for HSTs do not exploit features of HST communications such as the regular, repetitive, and predictable train route. This feature is considered in [4] for channel estimation in HSTs, which proposes to construct a database of detailed radio environment map (REM) for a given railway. The map should include various channel parameters at a given location along the HST rail, such as the number of paths, the amplitude, phase, delay, angle of arrival of every path, and so on. It can be initially constructed by offline field measurements and fine-tuned thereafter.

Inspired by [4], this paper investigates Doppler shift estimation in HSTs based on the REM, considering the dominant LOS path channel, which is mostly observed in HSTs [5]. REM could provide valuable information for Doppler shift estimation, such as the measured Doppler shifts and the corresponding speeds at each given location along the railway. Thus, at a certain location, given the carrier frequency, the measured speed, and location, the real Doppler shift can be estimated from the corresponding Doppler shift stored in REM (REME). However, the performance of REME is limited by the measurement errors, which are inevitable in practice. To improve the estimation performance, a maximum *a posteriori* estimator (MAPE) is proposed, which takes the estimation of REME as *a priori* knowledge and exploits the cyclic prefix (CP) structure of OFDM to provide a maximum *a posteriori* (MAP) estimation. The Cramer–Rao lower bounds (CRLBs) are derived, which can theoretically explain why MAPE is better than the classical CP-based estimator (CPE) [6]. It is shown by simulations that compared to existing estimators, MAPE can significantly improve the performance and reliability.

The rest of this paper is organized as follows. Section II describes the system model including the wireless channel model in HSTs. The MAPE algorithm is presented in Section III, following the description of REME. The CRLBs are also derived in Section III. Simulations are provided in Section IV. Finally, conclusions are drawn in Section V.

II. SYSTEM MODEL

As shown in Fig. 1, we consider an HST communication system, where land base stations (LBSs) are installed along the railway and communicate with mobile stations (MSs) in HSTs via LTE-R. A hierarchical two-hop network architecture is employed, where an access point (AP) is installed on the train to relay the signals between LBSs and MSs. There are two types of links in such a system: one from the LBS to the AP and the other from the AP to MSs in the carriage.

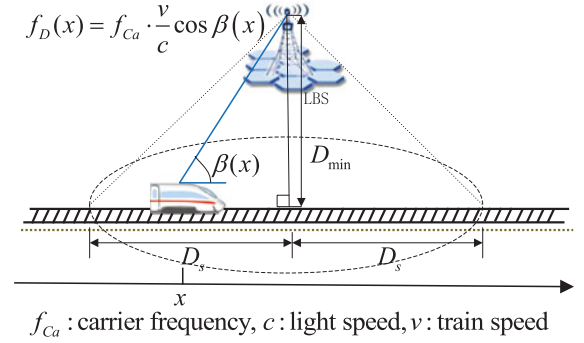


Fig. 1. Doppler shifts in HSTs.

Since the links between the AP and MSs are almost static, the focus is on the link from an LBS to the AP and the Doppler shift estimation at the AP.

In Third-Generation Partnership Project (3GPP) LTE specifications, the wireless channel of HSTs is modeled as an additive white Gaussian noise (AWGN) channel with changing Doppler shifts given by [1]

$$f_D(x) = f_{Ca} \cdot \frac{v}{c} \cos \beta(x) \quad (1)$$

where x is the location of the train, f_{Ca} is the carrier frequency, v represents the speed of the train, c denotes the speed of light, and $\beta(x)$ is the direction of arrival of an LOS path at the location x . It can be regarded as a simplified LOS model without fading and multipaths, which is referred to a LOS AWGN channel in this paper for simplicity. Although the Doppler shift changes rapidly with time in HSTs (maximum 4.9 kHz/s [1]), it can still be approximated by a constant value within the time interval of one OFDM symbol (the equivalent rate of change is 0.34 Hz per symbol). Hence, the Doppler shift can be treated as a fixed value f_m within one OFDM symbol in LTE-R.

However, the LOS AWGN channel is quite ideal. According to the field tests conducted on HSTs between Beijing and Tianjin in China [5], where 86% of the railway are operated on elevated lines, in addition to the strong LOS path, multiple paths exist most of the time. It has also been observed that the Doppler spread is not significant in such a system, and there is mainly one Doppler shift for the strong LOS path. Therefore, it is more reasonable to model the HST wireless channel as a multipath channel combining the LOS path channel in [7] with the 3GPP's time-varying Doppler shift curve [1]. Using this channel model, the received signal at the k th sample of the m th OFDM symbol in the downlink of LTE-R is given by

$$r_m(k) = \sqrt{P_{\text{LOS}}} \exp \left[j \left(\frac{2\pi \varepsilon k}{N} + \theta_0 \right) \right] s_m(k) + \sum_{l=0}^{L_c-1} h_l(k) \cdot s_m(k - d_l) + n_m(k) \quad (2)$$

where $k = 0, 1, \dots, N + L - 1$, L , and N are the numbers of samples of the CP and an effective OFDM symbol, respectively, $\varepsilon = f_m / \Delta f$ is the normalized Doppler shift, Δf is the subcarrier space (Hz), θ_0 represents an unknown initial phase, and L_c is the total number of uncorrelated paths. d_l is the delay of the l th path that is assumed to be a multiple of the sample duration T_s and $d_0 = 0$, $s_m(k)$ is the transmitted signal with zero mean and a power of σ_s^2 , and $n_m(k)$ is the AWGN noise with zero mean and a variance of σ_n^2 . Note that the path with zero delay is the LOS path, and its envelope follows the Rician distribution. Its Rician factor is $K = P_{\text{LOS}} / P_{\text{MP},0}$, where

P_{LOS} and $P_{\text{MP},0}$ are the powers of the LOS component and the disperse component $h_0(k)$, respectively. All of the $h_l(t)$ ($l = 0, 1, \dots, L_c - 1$) are assumed to follow the complex Gaussian distribution with zero mean and variance of $P_{\text{MP},l}$, so their envelopes follow the Rayleigh fading distribution. Their autocorrelation functions are given by $J_0(2\pi f_{d\max} t) \cdot P_{\text{MP},l}$, where $J_0(\cdot)$ is the zeroth-order Bessel function of the first kind, and $f_{d\max} = f_{\text{Ca}} \frac{v}{c}$ is the maximum Doppler shift. Without loss of generality, the average power of all the channel taps satisfies $P_{\text{LOS}} + \sum_{l=0}^{L_c-1} P_{\text{MP},l} = 1$. Note that when $(k - d_l) < 0$, the transmitted signal $s_m(k - d_l)$ should be in the previous OFDM symbol. For the purpose of clarity, the symbol index is not changed.

III. REME-BASED MEDIUM A POSTERIORI ESTIMATOR

A. REM Based Estimator

One important feature of HSTs is that the train travels along a constant route, which is regular, repetitive, and predictable, at a relatively stable speed. Thus, the Doppler shift experienced by the train at any location is also predictable. To exploit this feature in the Doppler shift estimation, a REM-based estimator, i.e., REME, could be constructed.

First of all, the REM can be obtained by offline field measurements using equipment with low errors (e.g., Doppler radar for Doppler shift measurements) [4], [8]. For any location x along the concerned route, the data stored in the REM, i.e., the measured Doppler shift $f_{D,\text{REM}}(x)$ of the LOS path, the corresponding speed $v_{\text{REM}}(x)$, and the carrier frequency $f_{\text{Ca},\text{REM}}$ of the measurement equipment are related to each other, as described in (1). Then, in practice, when an HST moves to the location x , the real Doppler shift it experiences on the LOS path $f_{D,\text{real}}(x)$, the current carrier frequency of LTE-R $f_{\text{Ca},\text{train}}$, and its speed $v_{\text{train}}(x)$ also follows the same relationship given by (1). Thus, $f_{D,\text{real}}(x)$ can be estimated from the data fetched from REM by

$$\hat{f}_{D,\text{est}}(x) = \frac{f_{\text{Ca},\text{train}} \cdot v_{\text{train}}(x)}{f_{\text{Ca},\text{REM}} \cdot v_{\text{REM}}(x)} f_{D,\text{REM}}(x). \quad (3)$$

Since $f_{\text{Ca},\text{train}}$ is usually a preknown system parameter, to carry out the estimation of Doppler shifts, the current location x of the train must be obtained first to fetch related information like $f_{\text{Ca},\text{REM}}$, $v_{\text{REM}}(x)$, and $f_{D,\text{REM}}(x)$ from the REM. The train speed $v_{\text{train}}(x)$ should be measured, which together with x can be generally obtained from a global positioning system, whose measurement error varies and can reach up to 100 m [9]. Note that REME is sensitive to the parameter measurement errors, especially the location errors. Due to lack of space, the detailed error analyses are omitted.

Note that (3) is not the only REME to estimate the Doppler shift. It can also be estimated by $\hat{f}_{D,\text{est}}(x) = f_{\text{Ca},\text{train}} \cdot v_{\text{train}}(x) \cdot \cos \beta(x)/c$ and used as the *a priori* information to construct a MAPE with CPE. Using $\beta(x)$ to compute the Doppler shift can avoid the storage of several quantities, i.e., $f_{\text{Ca},\text{REM}}$, $v_{\text{REM}}(x)$, and $f_{D,\text{REM}}(x)$. However, we found it very difficult to obtain a closed-form expression for the probability density function (PDF) of $\beta(x)$. Further investigation on this $\beta(x)$ -based MAPE is left for future research.

B. Derivation of MAPE

To further improve the estimation performance, a REME-based MAPE is proposed, which takes the Doppler shift estimations given by REME as *a priori* knowledge and exploits the CP structure of OFDM to provide a MAP estimation. Given the m th OFDM symbol with CP

and perfect time synchronization, MAPE for ε is designed by

$$\begin{aligned} \hat{\varepsilon} &= \arg \max_{\varepsilon} \{\Lambda(\varepsilon)\} \\ &= \arg \max_{\varepsilon} \{\log[f(r|\varepsilon, \hat{\varepsilon})] + \log[g(\hat{\varepsilon}|\varepsilon)] + \log[\xi(\varepsilon)]\} \end{aligned} \quad (4)$$

where $\Lambda(\varepsilon) = f(\varepsilon|r, \hat{\varepsilon})$ is the posterior distribution of the target Doppler shift ε , $\log\{\cdot\}$ is natural logarithm, $r = [r_m(0), r_m(1), \dots, r_m(N+L-1)]$ is the received signal vector, $\hat{\varepsilon}$ is the normalized Doppler shift provided by REME, $f(r|\varepsilon, \hat{\varepsilon})$ is the conditional PDF of the received signal, $g(\hat{\varepsilon}|\varepsilon)$ is the PDF of $\hat{\varepsilon}$ computed by REME in the condition that ε is given, and $\xi(\varepsilon)$ is the prior PDF of ε .

Note that $\log\{f(r|\varepsilon, \hat{\varepsilon})\} = \log\{f(r|\varepsilon)\}$ since the received signal r is independent of the Doppler shift $\hat{\varepsilon}$ estimated by REME. Different to the likelihood function of CPE in an AWGN channel, where the parameter ρ is independent of k , the likelihood function in a multipath channel is derived as

$$\begin{aligned} \log\{f(r|\varepsilon)\} &= - \sum_{k=0}^{L-1} \log(1 - |\rho(k)|^2) \\ &+ \frac{2}{(\sigma_s^2 + \sigma_n^2)} \sum_{k=0}^{L-1} \frac{\text{Re}(\rho^*(k) r_m^*(k+N) r_m(k))}{(1 - |\rho(k)|^2)} \\ &- \frac{1}{(\sigma_s^2 + \sigma_n^2)} \sum_{k=0}^{L-1} |\rho(k)|^2 (|r_m(k)|^2 + |r_m(k+N)|^2) \end{aligned} \quad (5)$$

where $\text{Re}(\cdot)$ is the real part of a complex number, $\rho^*(k)$ denotes the conjugation of $\rho(k)$, which given by

$$\begin{aligned} \rho(k) &= \frac{E[r_m(k) r_m^*(k+N)]}{\sqrt{E[|r_m(k)|^2] E[|r_m(k+N)|^2]}} \\ &= \frac{\sigma_s^2}{\sigma_s^2 + \sigma_n^2} [P_{\text{LOS}} \cdot \exp(-j2\pi\varepsilon) \\ &+ J_0(2\pi f_{d\max} N T_s) \cdot \Gamma(k)] \end{aligned} \quad (6)$$

where the channel tap power $\Gamma(k)$ is given by

$$\Gamma(k) = \begin{cases} \sum_{l=0}^{l'} P_{\text{MP},l}, & 0 \leq k < d_{L_c-1} \\ \sum_{l=0}^{L_c-1-k} P_{\text{MP},l}, & d_{L_c-1} \leq k < L \end{cases} \quad (7)$$

where $l' = \arg \max_l \{l|d_l < k\}$ is the channel tap index with the largest time delay among those taps with $d_l \leq k$.

For $\log\{g(\hat{\varepsilon}|\varepsilon)\}$ in (4), the PDF of $\hat{\varepsilon}$ is mainly related to the distributions of the measured train location and the Doppler shifts recorded in the REM. This paper focuses on the measured Doppler shift errors at each location, and the impacts of location errors on REME can be evaluated independently. Note that the Doppler shift of the LOS path at a certain location with a given speed is a constant value $\hat{\varepsilon}$. The measurement of this Doppler shift can be conducted by equipment with low errors, which are specially designed to get an accurate value of Doppler shift using directional narrow band signal. Usually, multiple measurements are needed to get the final estimation so that random errors can be averaged out [8]. Thus, the stored Doppler shift at a given location in REM can be modeled as a Gaussian distributed [4] variable with a mean of $\hat{\varepsilon}$ and a standard

deviation of $\tilde{\sigma}_m$ estimated from the data in REM. However, $\tilde{\varepsilon}$ is unknown in practice. So we approximate $\tilde{\varepsilon}$ by ε . Thus, $\log \{g(\tilde{\varepsilon}|\varepsilon)\}$ is given by

$$\log \{g(\tilde{\varepsilon}|\varepsilon)\} = \log \left\{ \frac{1}{\tilde{\sigma}_m \sqrt{2\pi}} \right\} - \frac{(\tilde{\varepsilon} - \varepsilon)^2}{2\tilde{\sigma}_m^2}. \quad (8)$$

The third item $\log \{\xi(\varepsilon)\}$ in (4) is the natural logarithm of the prior PDF of real Doppler shift, which is unknown in practice. Due to lack of accurate information on $\xi(\varepsilon)$, this term will not be exploited in the estimation to avoid an extra error.

After inserting (5) and (8) into (4), MAPE for ε is

$$\begin{aligned} \hat{\varepsilon} &= \arg \max_{\varepsilon} \{\Lambda(\varepsilon)\} \\ &= \arg \max_{\varepsilon} \left\{ -\frac{(\tilde{\varepsilon} - \varepsilon)^2}{2\tilde{\sigma}_m^2} - \sum_{k=0}^{L-1} \log(1 - |\rho(k)|^2) \right. \\ &\quad + \frac{2}{(\sigma_s^2 + \sigma_n^2)} \sum_{k=0}^{L-1} \frac{\text{Re}(\rho^*(k) r_m^*(k+N) r_m(k))}{(1 - |\rho(k)|^2)} \\ &\quad \left. - \frac{1}{(\sigma_s^2 + \sigma_n^2)} \sum_{k=0}^{L-1} |\rho(k)|^2 (|r_m(k)|^2 + |r_m(k+N)|^2) \right\}. \quad (9) \end{aligned}$$

C. Simplified MAPE in Strong LOS Channels

In HST scenarios with a strong LOS path, i.e., $P_{\text{LOS}} \gg \sum_{l=0}^{L_c-1} P_{\text{MP},l}$, MAPE can be simplified. Since $|J_0(x)| \leq 1$ holds for any real number x , it can be obtained that

$$\begin{aligned} |J_0(2\pi f_{d \max} N T_s) \cdot \Gamma(k)| &\leq \Gamma(k) \leq \sum_{l=0}^{L_c-1} P_{\text{MP},l} \quad (10) \\ &\ll |P_{\text{LOS}} \cdot \exp(-j2\pi\varepsilon)| = P_{\text{LOS}}. \end{aligned}$$

Thus, $\rho(k)$ in (6) can be approximated by

$$\rho = \rho(k) \approx \frac{P_{\text{LOS}} \cdot \sigma_s^2}{\sigma_s^2 + \sigma_n^2} \exp\{-j2\pi\varepsilon\} \quad (11)$$

which is independent with the sample index k . Thus, a simplified MAPE in LOS channels is given by

$$\begin{aligned} \hat{\varepsilon} &= \arg \max_{\varepsilon} \{\Lambda(\varepsilon)\} \\ &\approx \arg \max_{\varepsilon} \left\{ \alpha(\tilde{\sigma}_m)(\tilde{\varepsilon} - \varepsilon)^2 + \beta(L, \gamma, P_{\text{LOS}}) \right. \\ &\quad \left. \times \cos \left[2\pi\varepsilon + \arg \left\{ \sum_{k=0}^{L-1} (r_m^*(k+N) r_m(k)) \right\} \right] \right\} \quad (12) \end{aligned}$$

where $\alpha(\tilde{\sigma}_m) = -\frac{1}{2\tilde{\sigma}_m^2}$, and

$$\begin{aligned} \beta(L, \gamma, P_{\text{LOS}}) &= \frac{2P_{\text{LOS}} \cdot \gamma \cdot (\gamma + 1)}{[(1 - P_{\text{LOS}}^2) \cdot \gamma^2 + 2\gamma + 1] \cdot (\sigma_s^2 + \sigma_n^2)} \\ &\quad \times \left| \sum_{k=0}^{L-1} (r_m^*(k+N) r_m(k)) \right| \quad (13) \end{aligned}$$

TABLE I
COMPLEXITY COMPARISONS

	Number of complex additions	Number of complex multiplications
MAPE	$N_s(L+2)$	$N_s(L+9)$
CPE	$L-1$	$L+1$

where $\gamma = \sigma_s^2/\sigma_n^2$ is the SNR of the received signal. It can be seen from (12) that the first term $(\tilde{\varepsilon} - \varepsilon)^2$ is the squared residual Doppler shift error of REME weighted by $\alpha(\tilde{\sigma}_m)$. The smaller the variance $\tilde{\sigma}_m^2$ of the REME estimation $\tilde{\varepsilon}$, the larger the weight $\alpha(\tilde{\sigma}_m)$ of $(\tilde{\varepsilon} - \varepsilon)^2$, and the larger the impact of REME estimation error on $\Lambda(\varepsilon)$. On the other hand, the second term of (12) stands for the result obtained from the CPE weighted by $\tilde{\beta}(L, \gamma, P_{\text{LOS}})$ with $E\{\tilde{\beta}(L, \gamma, P_{\text{LOS}})\} \approx L(\gamma + 1)P_{\text{LOS}}$ when $\gamma \gg 1$. Therefore, as the length of CP, SNR, and the LOS path strength P_{LOS} increase, CPE becomes more reliable and its result should play a more important role in the MAPE.

It should be noted that this paper mainly focus on the Doppler shift estimation with a LOS channel in HST. The scenario with a non-LOS channel can be solved by existing methods [10]. Moreover, although MAPE is derived assuming one OFDM symbol m , it can be easily extended to multiple symbols as long as the variation of the Doppler shift and the channel can be neglected in that duration.

D. CRLBs

CRLB is the performance bound indicating the best estimation performance, depending on the observations and providing the lower bound on the variance of any unbiased estimate. CRLBs for joint CP- and REM-based estimation (e.g., MAPE) and CP-alone-based estimation (e.g., CPE) can be derived as $\text{Var}(\varepsilon) \geq \frac{1}{I_{\text{CP, REM}}(\varepsilon)}$ and $\text{Var}(\varepsilon) \geq \frac{1}{I_{\text{CP}}(\varepsilon)}$, respectively, where $I_{\text{CP, REM}}(\varepsilon) = I_{\text{CP}}(\varepsilon) + I_{\text{Prior}}(\varepsilon)$, and $I_{\text{Prior}}(\varepsilon) = \frac{1}{\sigma_{\tilde{\varepsilon}}^2}$. It can be seen that, theoretically, MAPE could provide better performance than CPE with a lower CRLB. $I_{\text{CP}}(\varepsilon)$ is

$$I_{\text{CP}}(\varepsilon) \approx 8\pi^2 \sum_{k=0}^{L-1} \left\{ \frac{1}{\frac{1}{|\rho(k)|^2} - 1} \right\}. \quad (14)$$

When an HST wireless channel with a strong LOS path is considered, using the simplified $\rho(k)$ in (11), $I_{\text{ML}}(\varepsilon)$ is

$$I_{\text{CP}}(\varepsilon) \approx 8\pi^2 L \frac{P_{\text{LOS}}^2 \cdot \gamma^2}{(1 - P_{\text{LOS}}^2) \cdot \gamma^2 + 2 \cdot \gamma + 1}. \quad (15)$$

It can be seen that the estimation performance can be improved by increasing CP length L (or equally the samples used in the ML estimator), SNR γ , and the LOS component power P_{LOS} .

E. Complexity Comparisons

In MAPE, to find the optimal solution of (12) and (13), a grid search method is employed. Since only a normalized Doppler shift ε needs to be searched, it is a 1-D search process. By exploiting prior information given by REME and CPE, the search range can be limited to $[\min(\tilde{\varepsilon}, \bar{\varepsilon}) - 3\tilde{\sigma}_m, \max(\tilde{\varepsilon}, \bar{\varepsilon}) + 3\tilde{\sigma}_m]$ wherein $\bar{\varepsilon}$ is the normalized Doppler shift given by CPE, and $\tilde{\varepsilon}$ and $\tilde{\sigma}_m$ are the normalized Doppler shift and standard deviation given by REME, respectively. Given the number of steps as N_s , the corresponding search step size is $p_{\text{step}} = \frac{\sigma_{\tilde{\varepsilon}} + |\tilde{\varepsilon} - \bar{\varepsilon}|}{N_s}$, wherein $|\cdot|$ is the absolute value. The complexity of MAPE and CPE is compared in Table I. It can be seen that the computational complexity of MAPE is N_s times higher than that of CPE. Since

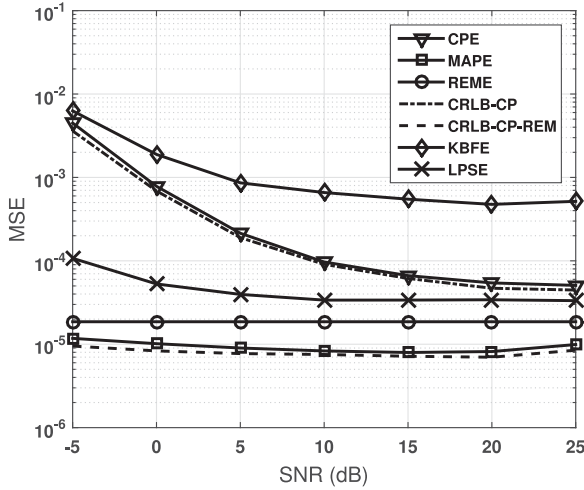


Fig. 2. MSE performance in HST channels.

a hierarchical two-hop network architecture has been employed in HSTs, an AP is installed on the train to relay the signals between LBSs and MSs in the train. It is AP that directly communicates with the LBS, which can employ powerful hardware with strong computing capability. Therefore, it is feasible for LTE-R with AP to employ more complicated algorithms like MAPE to enhance the system performance.

IV. SIMULATIONS

Simulations are conducted to evaluate the performance of MAPE. The channel settings follow [5], and the system configurations follow the LTE-R system in [1]. All the Doppler shifts in the simulations are normalized by the subcarrier spacing of LTE-R, which is 15 kHz. The number of searching steps N_s is 1000. Unless otherwise indicated, the measurement error ranges of the location, carrier frequency, train speed, Doppler shifts, and SNR are set to be ± 10 m, 0.1 ppm, ± 1 m/s, $\pm 5\%$, and $\pm 5\%$, respectively.

The performance of MAPE is investigated and compared to those of REME, CPE, KBFE, and LPSE. To ensure a fair comparison, we consider the same amount of overhead for CP-based methods and pilot assisted methods. Regarding [3], the CP length is used as an approximation of the channel length.

Fig. 2 compares the mean square errors (MSEs) of different estimators in HST channels. CRLBs of CP-alone-based estimation and joint CP- and REM-based estimation are also shown. According to (6), (7), and (14), CRLBs are dependent on the total power of multipath channel taps, resulting in the floor of CRLBs. It is also shown that the MSEs of CPE and pilot-based estimators (i.e., KBFE and LPSE) improve as SNR increases, but error floors appear at high SNR due to the multipath interference. KBFE performs the worst, because it cannot track the fast changing of the rectangular Doppler shift curve in HSTs, which it assumes changes in the shape of a sine curve. LPSE performs much better than KBFE since it is designed for the practical HST channel; thus, it is more robust to the multipath interference and has a lower error floor. However, the performance of LPSE is limited by the estimation error of the channel length, which is approximated by the CP length. Note that LPSE is even better than the CRLB of CP alone based estimation because LPSE is a pilot-based estimator, which is not limited by the CRLB of CP-based estimators.

The performance of REME keeps stable since it is not related to channel conditions. For MAPE, the MSE slightly decreases with SNR and approaches the corresponding CRLB. For sufficiently high SNRs,

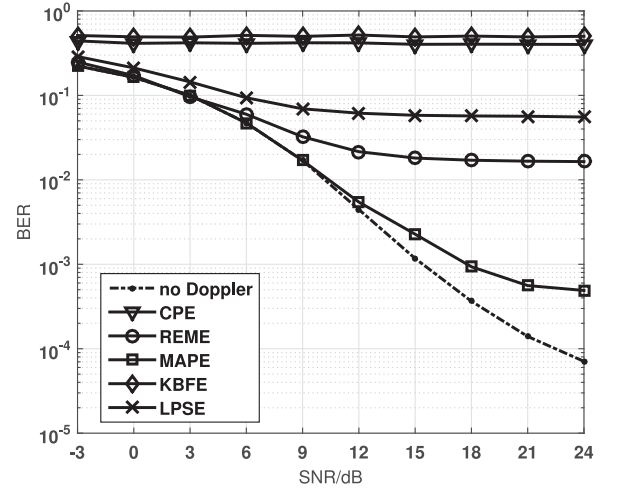


Fig. 3. BER performance in HST channels.

further increasing SNR slightly degrades the MSE of MAPE. This is because when SNR gets larger, the weight of CPE increases, whose estimation performance is limited by the error floor at high SNR. In summary, by exploiting the advantages of both REME and CPE, MAPE could provide better estimation than REME, CPE, KBFE, and LPSE in the practical HST channels.

Finally, Fig. 3 illustrates the system BER performance with different estimators in HST channels. Least-squares channel estimation and a simple frequency-domain equalizer are used, and no channel coding is applied. It can be seen that the BERs of CPE and KBFE are kept high as SNR increases. This is because their estimation performance is poor in the HST channel with multiple paths. Thus, the residual Doppler shift is still large after the compensation, and the receiver fails to recover the signals correctly. For LPSE, REME, and MAPE, the BER performance improves as SNR increases, but an error floor occurs when SNR is sufficiently high, and the better the Doppler estimation, the lower the error floor. This is because the BER is decided by both the background noise and the MSE of Doppler estimation. When SNR is low, the performance is dominated by the noise, and the BERs of LPSE, REME, and MAPE are close to each other. As SNR increases, the background noise reduces, and the BER is gradually dominated by MSE. As shown in Fig. 2, MAPE provides the best Doppler estimation, followed by REME and LPSE. The error floors in estimation result in the error floors in BERs. Thus, a significant performance gain in BER is achieved by MAPE over REME and LPSE.

V. CONCLUSION

In this paper, by exploring the unique feature of HST that the train moves along a given railway repetitively and predictably, a MAPE algorithm based on REME and the CP structure was proposed for Doppler shift estimation for LTE-R. Simulations verified that the proposed MAPE can provide better MSE performance in Doppler shift estimation and BER performance than CPE, REME, KBFE, and LPSE.

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Full-Duplex Device-to-Device-Aided Cooperative Nonorthogonal Multiple Access

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Abstract—This paper presents a full-duplex device-to-device (D2D)-aided cooperative nonorthogonal multiple access (NOMA) scheme to improve the outage performance of the NOMA-weak user in a NOMA user pair, where the NOMA-weak user is helped by the NOMA-strong user with the capability of full-duplex D2D communications. The expressions for the outage probability are derived to characterize the performance of the proposed scheme. The results show that the proposed cooperative NOMA scheme can achieve superior outage performance compared to the conventional NOMA and orthogonal multiple access (OMA). In order to further improve the outage performance, an adaptive multiple access (AMA) scheme is also studied, which dynamically switches between the proposed cooperative NOMA, conventional NOMA, and OMA schemes, according to the level of residual self-interference and the quality of links. The results show that the AMA scheme outperforms the above multiple access schemes in terms of outage performance.

Index Terms—Cooperative nonorthogonal multiple access (NOMA), device-to-device (D2D) communications, fifth-generation (5G) systems, full-duplex.

I. INTRODUCTION

Nonorthogonal multiple access (NOMA) has recently attracted significant attention both in academia and industry as a promising candidate technology in fifth-generation (5G) wireless networks [1], [2] and has already been introduced to fourth-generation long-term evolution-advanced systems as multiuser superposition transmission [3] to improve the downlink performance, due to its higher spectrum efficiency than traditional orthogonal multiple access (OMA). In order to further improve the performance of NOMA, recently cooperative NOMA (C-NOMA) schemes have attracted some attentions. In [4], a half-duplex user-aided N -timeslot cooperative NOMA scheme is proposed, and the results show that the proposed cooperative NOMA scheme could improve the outage probability, due to the achievement of the maximum diversity gain. In [5], a novel cooperative simultaneous wireless information and power transfer NOMA protocol was proposed, in which the NOMA-strong users that were close to the source act as energy harvesting relays to help the NOMA-weak users. In [6], the application of NOMA to the coordinated multiple point systems was studied. In [7], a cooperative beamforming NOMA scheme was proposed, which employed intrabeam superposition coding of a multiuser signal at the transmitter and the spatial filtering of interbeam interference followed by the intrabeam successive interference cancellation (SIC) at the terminal receiver. In addition, the performance of NOMA for relaying networks was also investigated in [8] and [9].

Device-to-device (D2D) communications enable the direct communications between users [10], and D2D users can also play as relays to cooperate data transmission, which can improve the system performance by sharing with the bandwidth used by cellular base stations (BSs). With the help of full-duplex radio [11], full-duplex D2D communications can further improve the system performance in heterogeneous networks [12].

In order to further improve the outage performance of the NOMA-weak user¹ in a user pair and reduce cooperative delay, in this paper, we focus on full-duplex D2D-aided cooperative NOMA, in which the user pair is predefined and configured and the NOMA-strong user has the capability of full-duplex D2D communications. With the help of full-duplex operation, the NOMA-strong user can receive data from the BS and forward the data to the NOMA-weak user over the same carrier frequency simultaneously. Compared to the existed works, the contributions of this paper are summarized as follows.

- 1) We present a full-duplex D2D-aided cooperative NOMA scheme to improve the transmission reliability of the NOMA-weak user, in which the NOMA-weak user is cooperated by the NOMA-strong user with the capability of full-duplex D2D communications.
- 2) We analyze the outage performance and derive the expressions for the proposed full-duplex D2D-aided cooperative NOMA scheme.

Manuscript received December 25, 2015; revised June 27, 2016; accepted July 29, 2016. Date of publication August 12, 2016; date of current version May 12, 2017. This work was supported in part by the National Natural Science Foundation of China under Grant 61571373; the National Science and Technology Major Project under Grant 2016ZX03001018-002; the National High-tech R&D Program of China under Grant 2014AA01A707; the Fundamental Research Funds for the Central Universities under Grant 2682016ZDPY06; Huawei HIRP Project under Grant YB201504; the EU Marie Curie Project "QUICK" under Grant 612652; and the Big Data Processing in Wireless Networks, Sweden STINT initiation grant. The work of Z. Zhang was also supported by the KTH-CSC Program. The review of this paper was coordinated by Dr. D. W. Matolak.

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Digital Object Identifier 10.1109/TVT.2016.2600102

¹Note that for the two users that are served in NOMA, the user with a higher signal-to-interference-plus-noise ratio (SINR) is considered as the NOMA-strong user, while the user with a lower SINR is considered as the NOMA-weak user.